Recall (multi-index notation):

$\alpha = (\alpha_1, \dots, \alpha_n)$, each $\alpha_i \geq 0$ is an integer, $|\alpha| = \alpha_1 + \dots + \alpha_n$

$u = u(x_1, \dots, x_n)$

$$D^\alpha u = \frac{\partial^\alpha u}{\partial x^\alpha} = \frac{\partial^{|\alpha|} u}{\partial x_1^{\alpha_1} \dots \partial x_n^{\alpha_n}}$$

Def: For an integer $k \geq 0$ and $0 < \gamma \leq 1$, we denote by $C^{k,\gamma}(\bar{U})$ the holder space of all functions $u : U \to \mathbb{R}$ that are k times continuously differentiable with all k^{th} order partial derivatives hölder continuous of exponent γ, with the norm

$$\|u\|_{C^{k,\gamma}(\bar{U})} = \sum_{|\alpha| \leq k} \|D^\alpha u\|_{C(\bar{U})} + \sum_{|\alpha| = k} [D^\alpha u]_{C^{0,\gamma}(\bar{U})}.$$

Thm: $C^{k,\gamma}(\bar{U})$ is a Banach space.

Ex:

$$\|u\|_{C^{2,\gamma}(\bar{U})} = \|u\|_{C(\bar{U})} + \sum_{i=1}^{n} \|u_{x_i}\|_{C(\bar{U})} + \sum_{i,j=1}^{n} \|u_{x_i x_i}\|_{C(\bar{U})}$$

$$+ \sum_{i,j=1}^{n} [u_{x_i x_j}]_{C^{0,\gamma}(\bar{U})}$$

$$[u]_{C^{0,\alpha}(\bar{U})} = \sup_{\substack{x,y\in U \\ x\neq y}} \frac{|u(x)-u(y)|}{|x-y|^{\alpha}}$$

$$\frac{|u(x)-u(y)|}{|x-y|^{\alpha}} \leq \|\nabla u\|_{C(\bar{U})} |x-y|^{1-\alpha}$$

$$|u(x)-u(y)| = \left| \int_0^1 \frac{d}{dt} u((1-t)x+ty)\, dt \right|$$

$$= \left| \int_0^1 \nabla u((1-t)x+ty)\cdot(y-x)\, dt \right|$$

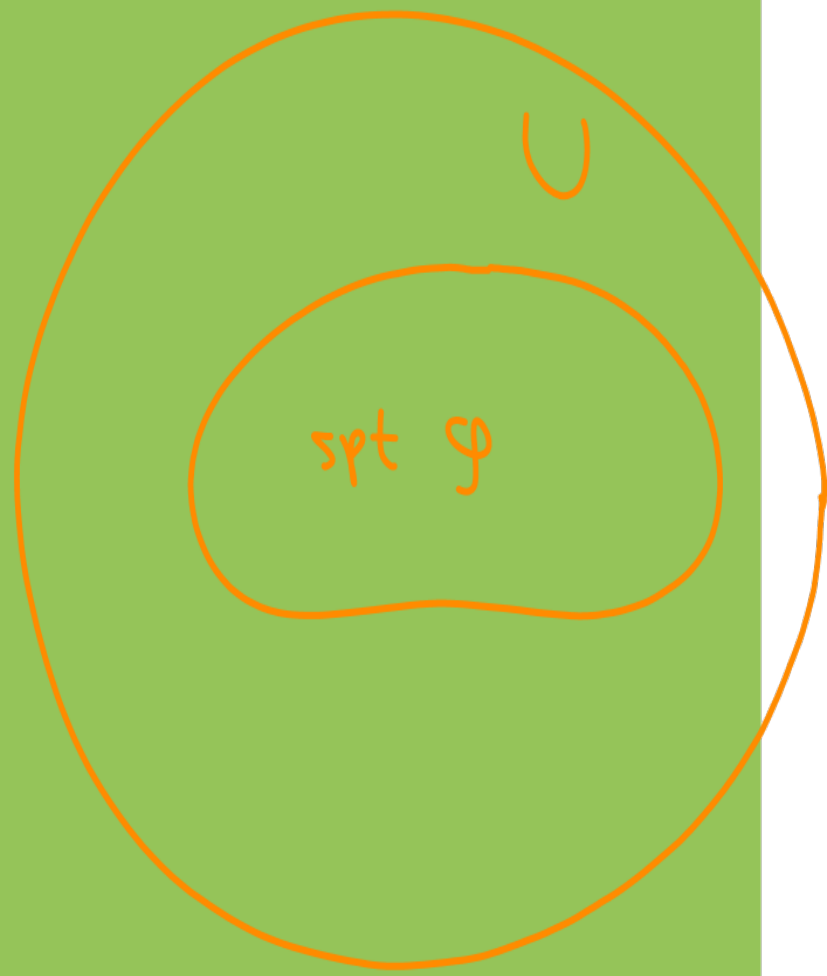
$$\leq \int_0^1 \overbrace{|\nabla u((1-t)x+ty)|}^{\leq \|\nabla u\|_{C(\bar{U})}} |x-y|\, dt \leq \|\nabla u\|_{C(\bar{U})} |x-y|$$

Sobolev Spaces

Def: Denote by $C_c^\infty(\mathbb{R}^n)$ the space of smooth functions with compact supports in $\mathbb{R}^n$. Functions in $C_c^\infty(\mathbb{R}^n)$ are called test functions.

Let $u \in C^k(\mathbb{R}^n)$ and let $\varphi \in C_c^\infty(\mathbb{R}^n)$. Then

$$\int_{\mathbb{R}^n} u_{x_i} \varphi \, dx = -\int_{\mathbb{R}^n} u \varphi_{x_i} \, dx.$$



Iterating this gives for any α with $|\alpha| \leq k$,

$$\int_{\mathbb{R}^n} \varphi D^\alpha u \, dx = (-1)^{|\alpha|} \int_{\mathbb{R}^n} u D^\alpha \varphi \, dx.$$

Recall (divergence thm):

$U \subset \mathbb{R}^n$ smooth bdd open set

$u, v \in C^1(\bar{U})$

$$\int_U u_{x_i} v \, dx = \int_{\partial U} u v \nu_i \, dS - \int_U u v_{x_i} \, dx$$

$\nu = (\nu_1, \dots, \nu_i)$ = outward unit normal to ∂U

<u>Recall:</u> $L^p(\mathbb{R}^n)$ = set of measurable functions $u : \mathbb{R}^n \to \mathbb{R}$

st $\displaystyle\int_{\mathbb{R}^n} |u|^p \, dx < \infty$

$L^p_{loc}(\mathbb{R}^n)$ = set of measurable functions $u : \mathbb{R}^n \to \mathbb{R}$

st $\displaystyle\int_V |u|^p \, dx < \infty$ for every open set $V \subset \mathbb{R}^n$

with $\bar{V}$ compact

If $u \in L^1_{loc}(\mathbb{R}^n)$, then

$$\left| \int_{\mathbb{R}^n} u D^\alpha \varphi \, dx \right| \leq \int_V |u| \underbrace{|D^\alpha \varphi|}_{} dx$$

bdd since $D^\alpha \varphi$ is continuous and vanishes outside the compact set $V :=$ support of φ

$$\leq C \int_V |u| dx < \infty.$$

Weak derivatives

Def: Let $u \in L^1_{loc}(\mathbb{R}^n)$. We say that a function $v \in L^1_{loc}(\mathbb{R}^n)$ is the αth weak partial derivative of u if

$$\int_{\mathbb{R}^n} u D^\alpha \varphi \, dx = (-1)^{|\alpha|} \int_{\mathbb{R}^n} v \varphi \, dx \qquad \forall \varphi \in C_0^\infty(\mathbb{R}^n).$$

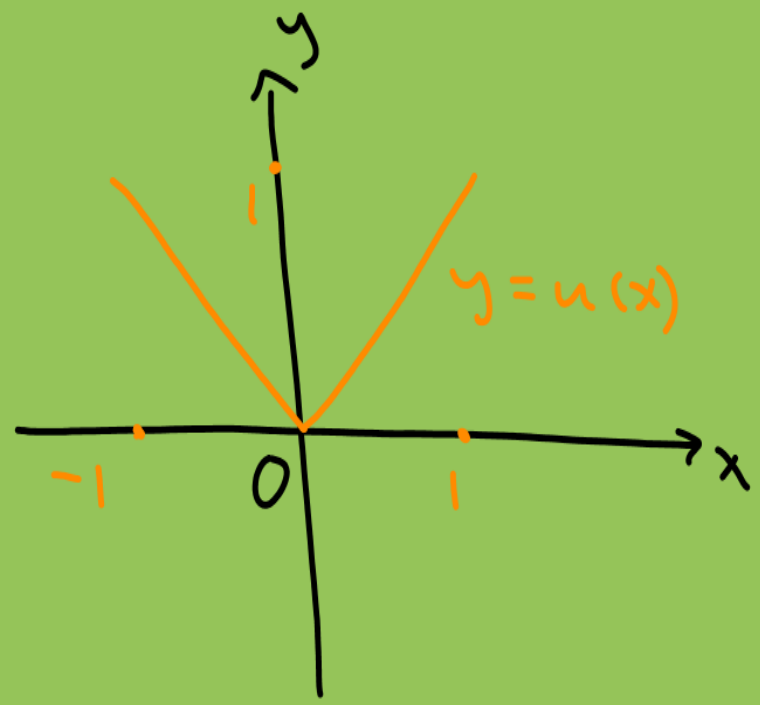
Let $U \subset \mathbb{R}^n$ be an open set and let $C_c^\infty(U)$ denote the space of smooth functions with compact supports in U.

<u>Def:</u> We say that $v = D^\alpha u$ weakly if

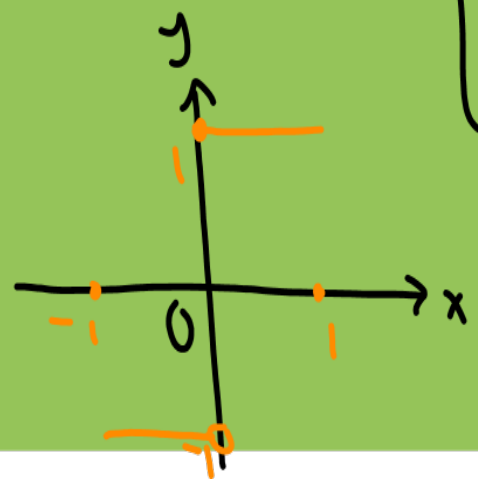
$$\int_U u D^\alpha \varphi \, dx = (-1)^{|\alpha|} \int_U v \varphi \, dx \quad \forall \varphi \in C_c^\infty(U).$$

Ex: $n=1$, $U=(-1,1)$, $u(x)=|x|=\begin{cases} x & \text{if } x\geq 0 \\ -x & \text{if } x<0 \end{cases}$

$u \notin C^1(\bar{U})$ since u is not differentiable at $x=0$



Let $v(x)=\begin{cases} 1 & \text{if } x\geq 0 \\ -1 & \text{if } x<0 \end{cases}$



Then $u' = v$ in the weak sense, i.e.,

$$\int_{-1}^{1} u(x)\varphi'(x)\,dx = -\int_{-1}^{1} v(x)\varphi(x)\,dx \qquad \forall \varphi \in C_c^\infty(-1,1)$$

<u>Pf</u>:

$$\int_{-1}^{1} u(x)\varphi'(x)\,dx = \int_{-1}^{0} -x\varphi'(x)\,dx + \int_{0}^{1} x\varphi'(x)\,dx$$

$$= \underbrace{-x\varphi(x)\Big|_{-1}^{0}}_{-\varphi(-1)} + \int_{-1}^{0}\varphi(x)\,dx + \underbrace{x\varphi(x)\Big|_{0}^{1}}_{\varphi(1)} - \int_{0}^{1}\varphi(x)\,dx$$

$$\int_{-1}^{1} u(x)\varphi'(x)\,dx = \underbrace{-\varphi(-1)+\varphi(1)}_{=0 \text{ since } \varphi \in C_c^\infty(-1,1)} + \underline{\int_{-1}^{0}\varphi(x)\,dx - \int_{0}^{1}\varphi(x)\,dx}$$

$$-\int_{-1}^{1} v(x)\varphi(x)\,dx = \underline{\int_{-1}^{0}\varphi(x)\,dx - \int_{0}^{1}\varphi(x)\,dx}$$

<u>Def:</u> Let $k \geq 0$ be an integer and let $1 \leq p \leq \infty$. The Sobolev space $W^{k,p}(U)$ consists of all $u \in L^1_{loc}(U)$ st for each α with $|\alpha| \leq k$, $D^\alpha u$ exists in the weak sense and belongs to $L^p(U)$, with the norm

$$= \begin{cases} \left(\sum_{|\alpha| \le k} \int_U |D^\alpha u| \, dx \right)^{1/p} & \text{if } 1 \le p < \infty \\ \sum_{|\alpha| \le k} \operatorname{ess\,sup} |D^\alpha u| & \text{if } p = \infty. \end{cases}$$

$$\|D^\alpha u\|^p_{L^p(U)}$$

$$\|D^\alpha u\|_{L^\infty(U)}$$

Ex: 1). $k = 0$

$$W^{0,p}(U) = L^p(U)$$

2). $k = 1$

$W^{1,p}(U)$ = space of functions $u \in L^p(U)$ st Du exists in the weak sense and belongs to $L^p(U)$

$$\|u\|_{W^{1,p}(U)} = \begin{cases} \left(\|u\|^p_{L^p(U)} + \sum_{i=1}^{n} \|u_{x_i}\|^p_{L^p(U)} \right)^{1/p} & \text{if } 1 \le p < \infty \\ \|u\|_{L^\infty(U)} + \sum_{i=1}^{n} \|u_{x_i}\|_{L^\infty(U)} & \text{if } p = \infty \end{cases}$$

Ex: $U = (-1,1)$, $u(x) = x$, $u'(x) = \begin{cases} 1 & \text{if } 0 \le x < 1 \\ -1 & \text{if } -1 < x < 0 \end{cases}$ weakly

bdd

u, u' bdd

$u \in W^{1,p}(U)$ for all $p \in [1, \infty]$